Практикум A4

145 marks, 50 blocks, nine techniques. The widest topic of section A and the only one where Paper 3 weighs more than Paper 2: 62 marks on Paper 1, 34 on Paper 2, 49 on Paper 3.

О практикуме

Тема взята целиком: 145 баллов, 50 блоков. Самая широкая в секции A и единственная в архиве, где Paper 3 весит больше Paper 2 (49 баллов против 34).

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-a4-polynomials

Собрано из ноутбука репозитория ib-math-aa-hl-past-papers. Проверки в PDF не работают — они для интерактивной версии.

Practicum A4: polynomials

145 marks, 50 blocks, nine techniques. The widest topic of section A and the only one where Paper 3 weighs more than Paper 2: 62 marks on Paper 1, 34 on Paper 2, 49 on Paper 3.

Material. 50 blocks and 145 marks from the AA HL archive, sessions May 2021 — November 2025. Every task below is a real past-paper question, cited.

The main idea of the topic. A polynomial can be read from two sides — by its coefficients and by its roots — and the whole topic consists of moving between those two readings.

- Coefficients → roots: the factor theorem, division, factorising.
- Roots → coefficients: Vieta's formulas, symmetric sums.

In nearly every question you have to work out which way to go. The sign is

simple: if the roots **do not have to be found** — «find the product of the solutions», «show that $a^2 = 3b$» — then the way is from the roots to the coefficients, and Vieta will save you a page.

How to work

1. Read the map of techniques. Techniques 1–3 are coefficients → roots, 4–8 are roots → coefficients, and 9 stands apart and leads into integration.
2. Work **on paper**. Long division and substitution into the factor theorem are hand work, and all the M1s are for them.
3. Enter the answer in whichever form suits you — except in the tasks where the question demands a particular **form**. There the check looks at the form too, and the task says so.
4. The last blocks are a recognition trainer and one question on a timer.

Difficulty marks: 🟢 the technique on its own · 🟡 the technique in a wrapper · 🔴 several techniques, or a whole exam question.

```
import sys
sys.path.append('..')          # from
practicum/number_algebra to practicum/kit/
import sympy as sp             # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
factor, apart, solve, sqrt, I and the rest

language('en')                 # this
notebook is in English, and so are the checks

a, b, c, d = symbols('a b c d')
m, n, p, q, r = symbols('m n p q r')
z, w = symbols('z w')

print('ready; sympy', sp.__version__)
print('factorise:               ',
factor(3*x**3 + 5*x**2 + x - 1))
print('divide with remainder:  ', div(2*x**2
- 5*x - 3, x - 1, x))
print('partial fractions:       ', apart(1/((x
+ 1)*(2*x + 1)), x))
print('roots numerically:       ', Poly(x**3 -
7*x**2 + 12*x + 1, x).nroots())
```

Map of techniques

#	Technique	Trigger in the question	First move
1	A factor and a remainder	«show that $(x-a)$ is a factor», «the remainder is …»	compute $P(a)$
2	Division with a remainder	«express in the form $Q(x)+\frac{R}{d(x)}$»	long division or matching
3	A repeated root	«divisible by $(x+1)^2$», «repeated root»	$P(a)=0$ **and** $P'(a)=0$
4	Vieta for a quadratic	«roots α,β», $\alpha^2+\beta^2$, the roots are not asked for	$\alpha+\beta=-\frac{b}{a}$, $\alpha\beta=\frac{c}{a}$
5	Vieta for a cubic and above	«write down the sum of the roots», a relation between the roots is given	the three symmetric sums
6	Complex roots in pairs	the coefficients are real, one root is imaginary	write down the conjugate

#	Technique	Trigger in the question	First move
7	Symmetric sums	$\alpha^2+\beta^2+\gamma^2$ in terms of p and q	$(\alpha+\beta+\gamma)^2$
8	An equation with new roots	«whose roots are $\frac{1}{\alpha^3}$», «$\frac{1}{z_i}$»	the substitution $w=\frac{1}{z}$, or Vieta
9	Partial fractions	«express in partial fractions»	the identity and convenient values of x

The ladder rests on two supports. Techniques 1–3 go from the coefficients to the roots: substitute — find a factor — divide — get the other roots. Techniques 4–8 go back the other way: the roots are not found at all, and you work with their sums and products, because those are expressed through the coefficients immediately.

Technique 9 stands apart. It says nothing about the roots, but it turns a fraction into a sum that can be integrated — and in the archive it is almost always part (a) of a long integration question.

What saves the most time. Before computing, look at whether the roots are asked for. Questions like «find the product of the solutions», «show that $a^2 - 3b = 0$», «find q and r» are solved without finding a single root. Whoever starts with the formula for the roots of a cubic loses not marks but time — and with it everything else.

Where the calculator is in this topic

Formally 83 marks out of 145 are on calculator papers, that is, 57%. That is exactly the middle of the archive and a figure that misleads.

	marks	of which really need a GDC
Paper 1	62	—
Paper 2	34	0
Paper 3	49	4

The real work for a calculator in this topic is worth **four marks**: running through the integers q in task 13(e) and

solving a system of three linear equations in one block from November 2022. Everything else is algebra, done the same way on either paper.

The reason is in how the topic is built. A calculator can find roots, and the questions here are about what is tied to the roots **without finding them**. The equation $x^3 - 7x^2 + qx + 1 = 0$ with the letter q a calculator will not solve, while Vieta does it in a line.

Where a GDC does help:

- **Check a factorisation.** Expand the product found and compare — ten seconds and three marks saved.
- **Run through integers.** «The smallest integer q for which …» — straight searching on a graph or in a table; the markscheme accepts that explicitly.
- **Find an integer root of a quartic.** Drawing the graph and seeing it cross the axis at $x = -1$ is faster than trying the divisors of the constant term.

Where it is useless and harmful: wherever there is a letter in the question. And in this topic a letter is almost everywhere.

Theory 1. A root, a factor, a remainder

The base of the whole left half of the topic is the **remainder theorem**:

$$\text{the remainder on dividing } P(x) \text{ by } (x-a) \;=\; P(a).$$

It is proved in one line. Write the division with a remainder: $P(x) = (x-a)Q(x) + R$, where R is a constant because the degree of the remainder is less than the degree of the divisor. Put $x = a$: the first term vanishes and $P(a) = R$ is left.

Hence the **factor theorem** — the same fact with $R = 0$:

$$(x-a) \text{ is a factor of } P(x) \iff P(a) = 0.$$

Careful with a factor that is not monic. For $(2x+1)$ the root is $-\frac{1}{2}$, not -1

and not -2. The general rule: the factor $(\alpha x+\beta)$ corresponds to $P\left(-\frac{\beta}{\alpha}\right)=0$. That loses marks more often than anything else in the theorem.

How it looks in a question. There are usually two conditions, and each gives an equation:

In the question	The equation
$(x+4)$ is a factor of f	$f(-4)=0$
the remainder is 7 when divided by $(x-2)$	$f(2)=7$
f is divisible by $(x-5)^2$	$f(5)=0$ **and** $f'(5)=0$

Two letters — two conditions — a linear system. There is nothing harder in this technique, and it brings 26 marks of the archive.

```
# A warm-up (not a task): the remainder
theorem on a live example.
f = 3*x**3 + 5*x**2 + x - 1
for a0 in [-2, -1, 0, 1, 2]:
    print(f'dividing by (x - {a0:2d}):
remainder {div(f, x - a0, x)[1]},'
          f'   f({a0:2d}) = {f.subs(x, a0)}')
```

Task 1 🟢 — a factor in one substitution

Given the polynomial $P(x) = 3x^3 + 5x^2 + x - 1$, show that $(x + 1)$ is a factor of $P(x)$.

Enter the value that shows it.

Source: May 2024 TZ1, Paper 1, Q11(a) (2 marks, no calculator).

```
my1 = ...                # the value that
turns the factor theorem into a check

check_num('Task 1', my1, 6, '5feceb66ffc8')
```

Task 2 🟢 — two conditions, two letters

Given $f(x) = 2x^4 - 6x^3 + px^2 + qx - 2$. It is known that $(x - 1)$ is a factor of $f(x)$ and that the remainder on dividing $f(x)$ by $(x - 3)$ is -2. Find p and q.

Source: May 2025 TZ3, Paper 2, Q2 (5 marks, calculator).

```
my2_p = ...
my2_q = ...

check_expr('Task 2, p', my2_p,
'2d9ab22507e5')
check_expr('Task 2, q', my2_q,
'2aeb14a25e82')
```

Task 3 🟡 — when the system has two solutions

Given $P(x) = 2x^2 + qx + r$. The remainder on dividing by $(x + 1)$ is 12, and the roots α and β of the polynomial are **real** and satisfy $\alpha^2 + \beta^2 = \dfrac{37}{4}$. Find q and r.

First enter **both** values of q that the system gives. Then the single one that survives the check of the condition, and the r that goes with it.

Source: November 2025 TZ3, Paper 2, Q6 (7 marks, calculator).

```
my3_all = [...]          # both roots of the
equation in q
my3_q = ...              # the one that
satisfies the question
my3_r = ...

check_set('Task 3, the candidates', my3_all,
'649d8f631f7e')
check_expr('Task 3, q', my3_q,
'd0cb37735b24')
check_expr('Task 3, r', my3_r,
'3b99ffe7f435')
```

Theory 2. Division and factorising

Division with a remainder for polynomials works as it does for integers:

$$P(x) = d(x) \\ Q(x) + R(x), \\ \deg R < \deg d.$$

The condition on the degree is not a formality but part of the definition. While the degree of the remainder is not less than the degree of the divisor, the division can go on and the writing is

simply unfinished. The check in this practicum watches for that separately.

Three ways to divide. All three are legitimate; the markscheme accepts any.

Long division. Slow, but it needs no guessing and always works.

Matching coefficients. Write $3x^2 + 7x - 2 = (x + 2)(Ax + B) + R$, expand, equate the coefficients and get $A = 3$, $B = 1$, $R = -4$. For division by a linear factor that is usually faster than long division.

Synthetic division. For division by $(x - a)$ — a table of one row.

Factorising is a different requirement already. «Express $P(x)$ as a product of three linear factors» asks not for a value but for a **form of writing**. The answer $3x^3 + 5x^2 + x - 1$ equals the right answer but earns no mark: there is no factorisation in it.

The order of work: find one root by substitution (the candidates are the divisors of the constant term divided by

the divisors of the leading one), divide, and factorise what is left by the ordinary school methods.

Task 4 🟢 — separate the whole part

Show that for $x \neq 1$

$$\frac{2x^2 - 5x - 3}{x - 1} = 2x - 3 - \frac{6}{x - 1}.$$

Enter the quotient and the remainder of the division of $2x^2 - 5x - 3$ by $(x - 1)$.

Source: November 2021, Paper 1, Q6(a) (2 marks, no calculator).

```
my4_quo = ...            # the quotient
my4_rem = ...            # the remainder

verify_division('Task 4', my4_quo, my4_rem,
2*x**2 - 5*x - 3, x - 1)
```

Task 5 🟢 — three linear factors

The same $P(x) = 3x^3 + 5x^2 + x - 1$ as in task 1. Express it as a product of three linear factors.

What is checked here is the **form of the writing**: an answer equal to $P(x)$ but not written as a product does not count.

Source: May 2024 TZ1, Paper 1, Q11(b) (3 marks, no calculator).

```
my5 = ...                # a product of three
brackets

verify_factored('Task 5', my5, 3*x**3 +
5*x**2 + x - 1, n=3)
```

Task 6 🟡 — the factor is given as a quadratic

The polynomial $P(x) = x^3 + ax^2 + bx + c$ is divisible by $x^2 + 3x + 2$. The numbers a, b, c are distinct and belong to $\{1, 2, 3, 4, 5\}$.

(i) Express b in terms of a. **(ii)** Show that only one set works. **(iii)** Factorise the resulting $P(x)$ into linear factors.

Source: November 2022, Paper 1, Q11(b) (9 marks, no calculator).

```
my6_b = ...              # an expression in
terms of a
my6_a, my6_bn, my6_cn = ..., ..., ...
my6_factored = ...       # a product of
linear factors

check_series('Task 6(i)', my6_b,
'b739173b8a1b', var=a)
check_num('Task 6(ii), a', my6_a, 6,
'4b227777d4dd')
check_num('Task 6(ii), b', my6_bn, 6,
'ef2d127de37b')
check_num('Task 6(ii), c', my6_cn, 6,
'd4735e3a265e')
verify_factored('Task 6(iii)', my6_factored,
x**3 + 4*x**2 + 5*x + 2, n=3)
```

Theory 3. A repeated root

If $(x-a)^2$ divides $P(x)$, then a is a root not only of P but of P'. It is checked by differentiating directly: from $P=(x-a)^2S(x)$ we get

$$P'(x)=2(x-a)S(x)+(x-a)^2S'(x),$$

and both terms contain $(x-a)$. Hence the working rule:

$$(x-a)^2 \mid P(x) \iff P(a) = 0 \text{ and } P'(a) = 0.$$

One condition has become two — exactly as many as are needed to find two letters.

The second route, with no derivative. Write $P(x) = (x-a)^2\left(x^2 + \alpha x + \beta\right)$, expand and equate coefficients. Longer, but it needs no differentiation and on Paper 1 is sometimes safer.

What a repeated root looks like on a graph. The graph touches the x-axis rather than crossing it. A multiplicity of 3 crosses it, but with a horizontal tangent. On Paper 2 that is a way of seeing the answer before any algebra.

Task 7 🟡 — divisible by the square of a bracket

Find p and q for which the polynomial

$$x^4 + px^3 - 2x^2 + qx - 3$$

is divisible by $(x+1)^2$.

The check here stores nothing: your values are substituted into the polynomial and the real remainder of the division is computed.

Source: May 2025 TZ2, Paper 1, Q7 (5 marks, no calculator).

```
my7_p = ...
my7_q = ...

verify_divisible('Task 7', x**4 + p*x**3 -
2*x**2 + q*x - 3, (x + 1)**2,
                 {p: my7_p, q: my7_q})
```

Theory 4. Vieta's formulas

The second half of the topic. Expand the brackets in a polynomial written through its roots:

$$\begin{aligned}&(x-\alpha)(x-\beta)(x-\gamma)\\ &= x^3\\ &\qquad -(\alpha+\beta+\gamma)x^2\\ &\qquad +(\alpha\beta+\beta\gamma+\gamma\alpha)x\\ &\qquad -\alpha\beta\gamma.\end{aligned}$$

Comparing with $x^3 + px^2 + qx + s$ gives three equalities at once. The general rule for $a_n x^n + \cdots + a_0$:

$$\begin{aligned} \sum \alpha_i &= -\frac{a_{n-1}}{a_n}, \\ \sum_{i<j} \alpha_i \alpha_j &= \frac{a_{n-2}}{a_n}, \\ &\cdots, \\ \prod \alpha_i &= (-1)^n \frac{a_0}{a_n}. \end{aligned}$$

The signs alternate. For a quadratic $ax^2 + bx + c$:

$$\alpha + \beta = -\frac{b}{a}, \qquad \alpha\beta = \frac{c}{a}.$$

Three places where marks are lost.

The leading coefficient is forgotten. For $3x^2 + 7x - 2$ the sum of the roots is $-\frac{7}{3}$, not -7. Dividing by a is compulsory.

The sign is confused. The sum carries a minus, and so does the product for a cubic. It is easier to remember through the alternation: $-, +, -, +$ starting from the sum.

The roots are hunted. If the question asks for $\alpha^2 + \beta^2$ or the product of the solutions, the roots are not needed. The identity

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

does the work, and in various wrappers it brings most of the marks of the technique.

Task 8 🟢 — the product of the roots with a letter

Write down an expression for the product of the roots of the equation

$$kx^2 - (k+3)x + 2k + 9 = 0$$

in terms of k.

Source: May 2022 TZ1, Paper 2, Q8(a) (1 mark, calculator).

```
my8 = ...                # the product of the
roots in terms of k

check_series('Task 8', my8, '84370cf7c1f6',
var=k)
```

Theory 5. An equation with new roots

The question sounds like this: an equation has roots α, β; write the equation whose roots are $\frac{1}{\alpha^3}$ and $\frac{1}{\beta^3}$ — **without finding α and β.**

Route one: substitution. If the new root is $w = f(z)$, express z in terms of w and substitute into the original equation. For $w = z + 2$ that is $z = w - 2$:

$$\begin{aligned} z^2 - 6z + 5 = 0 & \\ \Longrightarrow\ & (w-2)^2 - 6(w-2) + 5 \\ = 0 & \\ \Longrightarrow\ & w^2 - 10w + 21 \\ = 0. & \end{aligned}$$

Three lines, and it is the best way when the transformation can be inverted explicitly. Checking yourself is easy: the roots were 1 and 5, and they have become 3 and 7.

Route two: through Vieta. Compute the sum and the product of the new roots in terms of the old ones, then assemble

the equation $x^2 - Sx + P = 0$. It is needed when substitution leads to trouble — for example when the new roots are powers of the old ones:

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta, \qquad \alpha^2\beta^2 = (\alpha\beta)^2.$$

Both parts are expressed through the sum and the product, that is, through the coefficients, and no root had to be found. For higher powers the same thing works: the sum of powers is reduced to the sum and the product by a suitable identity, and it is exactly such an identity that the exam usually asks you to prove as a separate part — not for decoration, but because without it the second route does not go through.

The check in the notebook is built on the substance. There are no reference coefficients: the roots of your equation are computed numerically and compared with the transformed roots of the original. So an equation multiplied by any number passes too — its roots are

the same, and the markscheme accepts it as well.

Task 9 🟡 — an identity and a new equation

(a) Prove the identity $(p+q)^3 - 3pq(p+q) \equiv p^3 + q^3$. Enter the difference of the two expanded parts of the left-hand side.

(b) The equation $2x^2 - 5x + 1 = 0$ has roots α and β. Find m and n such that the roots of $x^2 + mx + n = 0$ are $\frac{1}{\alpha^3}$ and $\frac{1}{\beta^3}$. Finding α and β is not allowed.

Source: November 2021, Paper 2, Q6 (8 marks, calculator).

```
my9_a = ...              # (p+q)**3 - 3*p*q*
(p+q), both brackets expanded
my9_m = ...
my9_n = ...

check_series('Task 9(a)', my9_a,
'c7c28849e387', var=p)
verify_root_transform('Task 9(b)', [1, my9_m,
my9_n],
                      2*x**2 - 5*x + 1,
lambda t: 1/t**3)
```

Task 10 🟡 — a relation between the roots is given

The equation

$$x^3 - kx^2 + \tfrac{k^2}{2}\,x - 3k = 0, \qquad k > 0,$$

has roots α, β and $\alpha + \beta$.

First express $\alpha\beta$ in terms of k, then find k.

Source: May 2021 TZ2, Paper 1, Q7 (5 marks, no calculator). The question is given in a consistent form; the discrepancy with the archive's tags is dealt with in the solution.

```
my10_ab = ...            # alpha*beta in
terms of k
my10_k = ...             # the value of k

check_series('Task 10, alpha*beta', my10_ab,
'bbb8afd88279', var=k)
check_expr('Task 10, k', my10_k,
'1d8e9144eab5')
```

Theory 6. Complex roots come in pairs

If **all the coefficients** of a polynomial are real and z_0 is a root of it, then $\overline{z_0}$ is a root as well. The reason is one line: conjugation commutes with addition and multiplication, so

$$\overline{P(z_0)} = P(\overline{z_0}),$$

and if the left-hand side is zero, so is the right. The condition about real coefficients is essential: $z - i$ has only one root.

What follows immediately.

- Non-real roots always come in an even number.
- A cubic with real coefficients **always has a real root** — there are three of them and an even number of non-real ones. Another way of saying the same: the graph of a cubic goes to $-\infty$ and $+\infty$, so it crosses the axis.
- A pair $z_0, \overline{z_0}$ gives a real quadratic factor $z^2 - 2\,\mathrm{Re}(z_0)\,z + |z_0|^2$. For $1 \pm i$

that is $z^2 - 2z + 2$.

How this works together with Vieta. A pair of conjugate roots contributes $2\,\mathrm{Re}(z_0)$ to the sum — a real number — and $|z_0|^2 > 0$ to the product. The imaginary parts cancel and Vieta's equations stay real. That is exactly why such questions are solved with no complex working at all.

Task 11 🔴 — find the parameter and all the roots

The polynomial

$$P(z) = z^3 - \frac{36}{m}z^2 + mz - 4m,$$
$$m \in \mathbb{R},$$
$$m > 0,$$

has a factor $z - 3i$. Find m and all the roots of P.

Source: May 2023 TZ1, Paper 1, Q7 (6 marks, no calculator).

```
my11_m = ...
my11_roots = [...]       # all three roots

check_num('Task 11, m', my11_m, 6,
'19581e27de7c')
check_complex_set('Task 11, the roots',
my11_roots, '7c3cc706f5bd')
```

Task 12 🟡 — an equation for the reciprocal roots

The equation $z^4 + 2z^2 + 4 = 0$ has roots z_1, z_2, z_3, z_4. Find p, q, r such that the numbers $\frac{1}{z_1}, \ldots, \frac{1}{z_4}$ satisfy the equation

$$pw^4 + qw^2 + r = 0.$$

Source: May 2025 TZ2, Paper 1, Q12(d) (3 marks, no calculator).

```
my12_p, my12_q, my12_r = ..., ..., ...

verify_root_transform('Task 12', [my12_p, 0,
my12_q, 0, my12_r],
                      z**4 + 2*z**2 + 4,
lambda s: 1/s, var=z)
```

Theory 7. Symmetric sums and the reality of the roots

The prettiest construction in the topic, and all of it comes from one technique: **a symmetric combination of the roots is squared and Vieta's formulas are recognised in it.** The square of a sum always gives the same thing:

$$\left(\sum \alpha_i\right)^2 = \sum \alpha_i^2 + 2\sum_{i<j} \alpha_i\alpha_j,$$

and both sums on the right are coefficients. So the sum of the squares of the roots is expressed through the coefficients, and no root has to be found.

The same technique works not only for the roots themselves. Take the reciprocals for a cubic $x^3 + px^2 + qx + r$. By Vieta

$$\sum \frac{1}{\alpha} = \frac{q}{-r}, \qquad \sum_{i<j} \frac{1}{\alpha_i\alpha_j} = \frac{-p}{-r} = \frac{p}{r},$$

and the same squaring gives

$$\sum \frac{1}{\alpha^2} = \left(\frac{q}{-r}\right)^2 - 2\cdot\frac{p}{r} = \frac{q^2 - 2pr}{r^2}.$$

Check it on $x^3 - 6x^2 + 11x - 6$, whose roots are $1, 2, 3$: the formula gives $\frac{121-72}{36} = \frac{49}{36}$, and that is exactly $1 + \frac{1}{4} + \frac{1}{9}$.

Where this leads. If you square not the sum of the roots but their pairwise **differences** and add them, the left-hand side is a sum of squares — and for real numbers that is non-negative. On the right, after Vieta is substituted, an expression in two coefficients is left. The result is a criterion that can be checked in your head and that shows that the roots are not all real. Deriving it is the work of task 13, and it is worth doing rather than reading.

What the criterion does not give. The converse is false: the opposite inequality does **not** imply that all the roots are real. The sum of the squares of the differences is then non-negative, but it is a complex sum, and its non-negativity guarantees nothing. The counterexample is task 13 as well.

For a polynomial of the fourth degree the same identity works word for word, but there are four squares in it rather than three, and the constant in the criterion changes with their number. Carrying the number over from the cubic case to the quartic one is the most frequent mistake in this question.

Task 13 🔴 — the Paper 3 investigation, the cubic case

Let α, β, γ be the roots of $x^3 + px^2 + qx + r = 0$.

(b)(i) Express $\alpha^2 + \beta^2 + \gamma^2$ in terms of p and q. **(b)(ii)** Express $(\alpha - \beta)^2 + (\beta - \gamma)^2 + (\gamma - \alpha)^2$ in terms of p and q. **(d)** For $x^3 - 7x^2 + qx + 1 = 0$ with $q = 17$, enter p^2 and $3q$. **(e)(i)** Find the smallest integer $q > 0$ for which $p^2 \geq 3q$ and yet the roots are still not all real.

Source: May 2022 TZ2, Paper 3, Q2(a)–(e) (15 marks, calculator).

```
my13_sq = ...            #
alpha^2+beta^2+gamma^2 in terms of p and q
my13_dif = ...           # the sum of the
squares of the differences, in p and q
my13_p2 = ...            # p^2 for x^3 - 7x^2
+ 17x + 1
my13_3q = ...            # 3q for the same
one
my13_q = ...             # the smallest
integer q from part (e)

check_series('Task 13(b)(i)',  my13_sq,
'a6387ceaabb3', var=p)
check_series('Task 13(b)(ii)', my13_dif,
'ad12273ee92f', var=p)
check_num('Task 13(d), p^2', my13_p2, 6,
'0e17daca5f3e')
check_num('Task 13(d), 3q', my13_3q, 6,
'031b4af5197e')
check_num('Task 13(e)', my13_q, 6,
'6b51d431df5d')
```

```
# Part (e) is the one place where a
calculator is genuinely needed.
# The condition p^2 >= 3q at p = -7 gives q
<= 16; we look at where the roots
# stop being all real.
for q0 in range(1, 17):
    found = Poly(x**3 - 7*x**2 + q0*x + 1,
x).nroots()
    n_real = sum(1 for s in found if
abs(im(s)) < 1e-9)
    print(f'q = {q0:2d}:  real roots
{n_real}',
          '  <-- counterexample' if n_real <
3 else '')
```

Task 14 🔴 — the Paper 3 investigation, the fourth degree

A continuation of the same question. Now $\alpha, \beta, \gamma, \delta$ are the roots of $x^4 + px^3 + qx^2 + \cdots = 0$.

(f)(i) Express $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ in terms of p and q. **(g)** For $x^4 - 2x^3 + 3x^2 - 4x + 5 = 0$, enter p^2 and $2q$. **(h)(ii)** Write down the integer root of $x^4 - 9x^3 + 24x^2 + 22x - 12 = 0$. **(h)(iii)** Divide that polynomial by $(x + 1)$ and enter the quotient and the remainder; then p^2 and $3q$ for the cubic factor obtained.

Source: May 2022 TZ2, Paper 3, Q2(f)–(h) (12 marks, calculator).

```
my14_sq = ...            # the sum of the
squares of four roots, in p and q
my14_p2, my14_2q = ..., ...
my14_root = ...          # the integer root
of the quartic
my14_quo, my14_rem = ..., ...
my14_pc2, my14_3qc = ..., ...

QUART = x**4 - 9*x**3 + 24*x**2 + 22*x - 12

check_series('Task 14(f)(i)', my14_sq,
'a6387ceaabb3', var=p)
check_num('Task 14(g), p^2', my14_p2, 6,
'4b227777d4dd')
check_num('Task 14(g), 2q', my14_2q, 6,
'e7f6c011776e')
check_num('Task 14(h)(ii)', my14_root, 6,
'1bad6b8cf971')
verify_division('Task 14(h)(iii)', my14_quo,
my14_rem, QUART, x + 1)
check_num('Task 14(h)(iii), p^2', my14_pc2,
6, 'ad5736686512')
check_num('Task 14(h)(iii), 3q', my14_3qc, 6,
'37834f2f2576')
```

Theory 8. Partial fractions

A technique that says nothing about the roots, but without which half the integrals do not go through. A proper fraction with a factorisable denominator is split into a sum:

$$\frac{1}{(x-2)(3x+1)} = \frac{A}{x-2} + \frac{B}{3x+1}.$$

Multiply both sides by the denominator:

$$1 = A(3x+1) + B(x-2).$$

From there two routes, both worth full marks.

Substituting convenient values. $x = 2$ kills the second term: $1 = 7A$, hence $A = \frac{1}{7}$. $x = -\frac{1}{3}$ kills the first: $1 = -\frac{7}{3}B$, hence $B = -\frac{3}{7}$. Faster, safer, and with almost nowhere to go wrong.

Equating coefficients. $3A + B = 0$ and $A - 2B = 1$. Longer, but it works where there are no convenient values.

A repeated factor needs two terms. For $(x+1)^2$ in the denominator you write

$$\frac{A}{x+1} + \frac{B}{(x+1)^2},$$

not a single fraction. If you write a single one, the system turns out inconsistent — a reliable sign that a term has been forgotten.

The check in the notebook looks at the form. Every term has to be a partial fraction: a numerator with no x, a denominator that is a power of a single linear factor. The answer $\frac{1}{(x+1)(2x+1)}$ equals itself but is not a decomposition, and the check will reject it — exactly as an examiner would.

Task 15 🟢 — two decompositions in one line

(a) Decompose $\dfrac{1}{x(k-x)}$ into a sum of the form $\dfrac{a}{x}+\dfrac{b}{k-x}$.

(b) Decompose $\dfrac{1}{1-4v^2}$ into a sum of the form $\dfrac{A}{1-2v}+\dfrac{B}{1+2v}$.

Sources: May 2021 TZ1, Paper 2, Q12(a) (3 marks) and November 2025 TZ3, Paper 2, Q12(c) (2 marks), both with a calculator.

```
my15_a = ...              # a sum of two
fractions, in terms of k
my15_b = ...              # a sum of two
fractions, in terms of v

check_apart('Task 15(a)', my15_a, 1/(x*(k -
x)))
check_apart('Task 15(b)', my15_b, 1/(1 -
4*v**2), var=v)
```

Task 16 🟡 — a repeated factor in the denominator

Let $Q(x) = (x+1)(2x+1)$.

(c) Decompose $\dfrac{1}{Q(x)}$ into partial fractions.

(d) Decompose into partial fractions

$$\frac{1 - \dfrac{x}{Q(x)}}{(1+x)(2x+1)}.$$

In the second part the denominator, once tidied, turns out to be a square, and there will be four terms.

Source: May 2024 TZ1, Paper 1, Q11(c)–(d) (5 marks, no calculator). The right-hand side of part (d) is recorded with

errors in the archive's tags; the working is in the solution.

```
Q = (x + 1)*(2*x + 1)

my16_c = ...              # the decomposition
of 1/Q(x)
my16_d = ...              # the decomposition
of the expression from part (d)

check_apart('Task 16(c)', my16_c, 1/Q)
check_apart('Task 16(d)', my16_d, (1 -
x/Q)/((1 + x)*(2*x + 1)))
```

Trainer: name the technique in five seconds

Below are fifteen statements with no solutions. For each one write the code of the technique — and nothing else. Nothing has to be computed; the aim is for the first move to be found before the computing starts.

The codes: `factor` (a factor and a remainder) · `div` (division with a remainder) · `mult` (a repeated root) · `vieta2` (Vieta, a quadratic) · `vieta3` (Vieta, degree above two) · `conj` (complex roots

in pairs) · `sym` (symmetric sums) · `transform` (an equation with new roots) · `apart` (partial fractions)

1. Show that $(x-2)$ is a factor of $x^3 - 3x^2 + 4x - 4$.
2. The equation $3x^2 + 7x - 2 = 0$ has roots α, β. Find $\alpha^2 + \beta^2$.
3. Write $\dfrac{5}{(x-1)(x+4)}$ as a sum of two fractions.
4. For which a does the polynomial $x^3 + ax + 2$ have a repeated root?
5. The roots of $x^3 + px^2 + qx + r$ are real. What can be said about p^2 and $3q$?
6. The equation with roots $\alpha + 1$ and $\beta + 1$, where α, β are the roots of a given one.
7. Write $\dfrac{x^3 + 1}{x - 2}$ in the form $Q(x) + \dfrac{R}{x-2}$.
8. A polynomial with real coefficients has the root $2 - 5i$. What else is known?
9. Write down the sum of the roots of $4x^5 - 3x^4 + x - 7 = 0$.

0. The remainder on dividing $f(x)$ by $(x-3)$ is 5. Find $f(3)$.
1. Decompose $\dfrac{2x+1}{x^2(x-3)}$ into partial fractions.
2. For $x^4 + px^3 + qx^2 + \dots$ express $\sum \alpha_i^2$ in terms of p and q.
3. The graph of $y = P(x)$ touches the x-axis at $x = 4$. What does that give?
4. Find the product of the solutions of the equation $2x^2 = 5x - 1$.
5. The equation $z^3 - 1 = 0$ has the root $z = 1$. Find the others.

```
answers = {
    1: '',   2: '',   3: '',   4: '',   5:
'',
    6: '',   7: '',   8: '',   9: '',  10:
'',
   11: '',  12: '',  13: '',  14: '',  15:
'',
}

trigger_check(answers, {1: '41a093100bbb', 2:
'a40c1a3b397d', 3: '1d40973eccd8', 4:
'c0c75a8e79ad', 5: '0599b22abfb1', 6:
'aa214ea38326', 7: 'cd35a2426062', 8:
'd1a73570aa07', 9: '6223d7789180', 10:
'41a093100bbb', 11: '1d40973eccd8', 12:
'0599b22abfb1', 13: 'c0c75a8e79ad', 14:
'a40c1a3b397d', 15: 'd1a73570aa07'})
```

Task 17 🔴 — on the timer, 12 minutes

A whole Paper 1 question with no calculator. Seven marks, target time about 12 minutes.

The cubic polynomial $g(x) = 2x^3 - 7x^2 + dx - e$ has roots α, β, γ.

(a) Write down $\alpha + \beta + \gamma$.

(b) The same α, β, γ are roots of the polynomial $h(z) = 2z^5 - 11z^4 + rz^3 + sz^2 + tz - 20$, and $z = \rho + 3i$ is a root of it too. Show that $\rho = 1$.

(c) Given that $h\left(\frac{1}{2}\right) = 0$, that α, β are positive and real with $\alpha < \beta$, and that γ is rational, find $\alpha\beta$ and then α and β.

Source: May 2024 TZ2, Paper 1, Q7 (7 marks, no calculator).

```
my17_sum = ...           # alpha + beta +
gamma
my17_rho = ...           # the value of rho
my17_ab = ...            # alpha*beta
my17_alpha, my17_beta = ..., ...

check_expr('Task 17(a)', my17_sum,
'ca24699bc818')
check_num('Task 17(b)', my17_rho, 6,
'6b86b273ff34')
check_num('Task 17(c), alpha*beta', my17_ab,
6, 'd4735e3a265e')
check_num('Task 17(c), alpha', my17_alpha, 6,
'6b86b273ff34')
check_num('Task 17(c), beta', my17_beta, 6,
'd4735e3a265e')
```

Solutions

Below is a full working of every task with the markscheme's marking. Open them after you have solved on your own: the working explains not only the answer but what exactly the marks are given for.

Solution 1 — a factor in one substitution

By the factor theorem $(x+1)$ divides $P(x)$ if and only if $P(-1)=0$. Compute:

$$\begin{aligned} P(-1) &= 3(-1)^3 + 5(-1)^2 + (-1) - 1 \\ &= -3 + 5 - 1 - 1 \\ &= 0. \qquad \blacksquare \end{aligned}$$

```
my1 = 0
```

The marking. M1 for substituting $x=-1$ (the expression itself has to be visible, not just the result), A1 for the zero with the conclusion «therefore $(x+1)$ is a factor». Long division with a zero remainder is accepted as an equal.

About the wording. This is a «show that» question: the answer is printed in the question and the mark is for the working. Writing «$P(-1) = 0$, QED» with no computation scores zero. The concluding sentence at the end is compulsory: without it the computing is visible but the answer to the question asked is not.

Solution 2 — two conditions, two letters

$(x - 1)$ is a factor, so $f(1) = 0$:

$$2 - 6 + p + q - 2 = 0 \implies p + q = 6.$$

The remainder on dividing by $(x - 3)$ is -2, so $f(3) = -2$:

$$\begin{aligned} 2 \cdot 81 - 6 \cdot 27 + 9p + 3q - 2 &= -2 \\ &\implies 162 - 162 + 9p + 3q \\ &= 0 \\ &\implies 3p + q \\ &= 0. \end{aligned}$$

Subtract the first from the second: $2p = -6$, hence $p = -3$ and $q = 9$.

```
my2_p = -3
my2_q = 9
```

The marking. M1 for applying the factor theorem, A1 for the first equation, M1 for applying the remainder theorem, A1 for the second equation, A1 for both values. Five marks, and three of them are for the equations being written at all.

About the calculator. This is Paper 2 and the system is allowed to be solved on a GDC. But it still has to be written down by hand, and a system of two linear equations is solved in your head faster than it is typed in. The calculator belongs here as a check: substitute the p, q found and confirm that $f(1) = 0$.

Solution 3 — when the system has two solutions

The remainder on dividing by $(x+1)$ is $P(-1)$:

$$2 - q + r = 12 \implies r - q = 10.$$

By Vieta $\alpha + \beta = -\frac{q}{2}$ and $\alpha\beta = \frac{r}{2}$. Then

$$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=\frac{q^2}{4}-r=\frac{37}{4}.$$

Substitute $r=q+10$:

$$\begin{aligned}\frac{q^2}{4}-q-10&=\frac{37}{4}\\ &\Longrightarrow\ q^2-4q-40\\ &=37\\ &\Longrightarrow\ q^2-4q-77\\ &=0.\end{aligned}$$

Factorise: $(q-11)(q+7)=0$, hence $q=11$ or $q=-7$.

Do both work? Check the discriminant of $2x^2+qx+r$:

- $q=-7,\ \ r=3$: $D=49-24=25>0$, the roots 3 and $\frac{1}{2}$ are real.
- $q=11,\quad r=21$: $D=121-168=-47<0$, the roots are not real.

The question requires real roots, so $q=-7$, $r=3$ remain.

```
my3_all = [-7, 11]
my3_q = -7
my3_r = 3
```

The marking. M1 for the remainder theorem, A1 for $r - q = 10$, M1 for the identity $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$, A1 for the second equation, M1 for solving the system, A1A1 for q and r.

Why this task is marked 🟡. The algebra is simple and the trap is at the end: the quadratic in q gives two roots, and the second looks no worse than the first. It is discarded only by checking the discriminant. In the archive's own tags the second root is not mentioned at all — and that is the case where the working of the question is shorter than the truth about it.

A general rule. Whenever the system in the letters has turned out quadratic, compute both solutions and check each against the question. A condition «the roots are real», «$k > 0$», «$n \in \mathbb{Z}^+$» is not in the question for decoration — it is the selection.

Solution 4 — separate the whole part

Divide by long division. $2x^2 : x = 2x$; multiply $2x(x-1) = 2x^2 - 2x$; subtract:

$$2x^2 - 5x - 3 - (2x^2 - 2x) = -3x - 3.$$

Then $-3x : x = -3$; multiply $-3(x-1) = -3x + 3$; subtract:

$$-3x - 3 - (-3x + 3) = -6.$$

The quotient is $2x - 3$ and the remainder -6. So

$$\frac{2x^2 - 5x - 3}{x - 1} = 2x - 3 - \frac{6}{x - 1}. \qquad \blacksquare$$

```
my4_quo = 2*x - 3
my4_rem = -6
```

The marking. The markscheme comes at this block from the other side: M1 for bringing the right-hand side to a common denominator, A1 for the numerators matching. Both directions are accepted, and the second is sometimes faster — multiplying $2x - 3 - \frac{6}{x-1}$ by $(x-1)$ is easier than dividing.

Why separate the whole part at all. For the sake of integration. The fraction $\frac{2x^2-5x-3}{x-1}$ does not integrate as it is, while $2x-3-\frac{6}{x-1}$ integrates in a line. The same technique for the same reason stands at the start of half the questions in section E5.

About the degree of the remainder. The check in the notebook watches separately that the degree of the remainder is less than the degree of the divisor. The answer «quotient $2x$, remainder $-3x-3$» is arithmetically right — the sum recovers the dividend — but it is not a division: the dividing can go on.

Solution 5 — three linear factors

From task 1 we know that $(x+1)$ is a factor. Divide:

$$3x^3+5x^2+x-1=(x+1)\left(3x^2+2x-1\right).$$

The quadratic factorises the ordinary way. Look for two numbers with product $3 \cdot (-1) = -3$ and sum 2: they are 3 and -1.

$$\begin{aligned} 3x^2 + 2x - 1 &= 3x^2 + 3x - x - 1 \\ &= 3x(x+1) - (x+1) \\ &= (x+1)(3x-1). \end{aligned}$$

So

$$P(x) = (x+1)^2(3x-1). \qquad \blacksquare$$

```
my5 = (x + 1)**2*(3*x - 1)
```

The marking. M1 for dividing by $(x+1)$, A1 for the quadratic quotient, A1 for the final product. Three marks.

What matters here is that $(x+1)$ came out twice. The root -1 is repeated, and the graph of P touches the x-axis there rather than crossing it. Technique 3 deals with such cases from the other side — when the multiplicity is given in the question and the coefficients are what have to be found.

About the form of the answer. The question asks for **a product of three**

linear factors, and that is a requirement on the writing. The check `verify_factored` parses exactly what is in the cell and does not factorise the answer itself: `3*x**3 + 5*x**2 + x - 1` equals the right answer but gets ❌. That is exactly what an examiner would do.

It also shows how this topic differs from A3. There `check_series` accepted an unexpanded bracket, because the question asked for an expansion and numerically a bracket equals its expansion. Here the question asks for the reverse — and so does the check.

Solution 6 — the factor is given as a quadratic

(i) Factorise the divisor: $x^2 + 3x + 2 = (x + 1)(x + 2)$. So $P(-1) = 0$ and $P(-2) = 0$:

$$
\begin{aligned}
&-1 \\
&\quad +a \\
&\quad -b \\
&\quad +c \quad\;\; = 0, \\
&\quad -8 \\
&\quad +4a \\
&\quad -2b \\
&\quad +c \\
&= 0.
\end{aligned}
$$

Subtract the first from the second to remove c:

$$-7 + 3a - b = 0 \implies b = 3a - 7. \qquad \blacksquare$$

(ii) From the first equation $c = 1 - a + b = 1 - a + 3a - 7 = 2a - 6$. Run through $a \in \{1, \dots, 5\}$:

a	$b = 3a - 7$	$c = 2a - 6$	**does it work?**
1	-4	-4	no, outside the set
2	-1	-2	no, outside the set
3	2	0	no, $0 \notin \{1, \dots, 5\}$
4	5	2	**yes**
5	8	4	no, 8 is outside the set

The only set: $a = 4$, $b = 5$, $c = 2$, and all three are distinct. ■

(iii) $P(x) = x^3 + 4x^2 + 5x + 2 = (x^2 + 3x + 2)(x + 1) = (x + 1)^2(x + 2)$.

```
my6_b = 3*a - 7
my6_a, my6_bn, my6_cn = 4, 5, 2
my6_factored = (x + 1)**2*(x + 2)
```

The marking. (i): M1 for factorising the divisor, M1 for applying the factor theorem, A1A1 for the two equations, M1 for eliminating c, A1 for $b = 3a - 7$. (ii): M1 for the search with the conditions checked, A1 for uniqueness. (iii): A1.

Where marks are lost. In part (ii) the markscheme requires exactly that **all the cases are checked**: the answer « $a = 4$» with no table gets M0A0, because uniqueness has not been shown. That is the same kind of requirement as the R1 in A7 — the mark is for the completeness of the reasoning, not for the right number.

A small thing worth a mark. Checking that a, b, c really are **distinct**: the

question says «distinct». The row $a = 3$ falls away not because of a repeat but because $c = 0$ is not in the set — but both conditions have to be checked.

Solution 7 — divisible by the square of a bracket

Write $f(x) = x^4 + px^3 - 2x^2 + qx - 3$. Divisibility by $(x+1)^2$ means

$$f(-1) = 0 \quad \text{and} \quad f'(-1) = 0.$$

The first condition:

$$1 - p - 2 - q - 3 = 0 \implies p + q = -4.$$

Differentiate: $f'(x) = 4x^3 + 3px^2 - 4x + q$. The second condition:

$$-4 + 3p + 4 + q = 0 \implies 3p + q = 0.$$

Subtract: $2p = 4$, hence $p = 2$ and $q = -6$.

Check: $x^4 + 2x^3 - 2x^2 - 6x - 3 = (x+1)^2(x^2 - 3)$. ■

```
my7_p = 2
my7_q = -6
```

The marking. M1 for the condition $f(-1)=0$, A1 for the first equation, M1 for differentiating and $f'(-1)=0$, A1 for the second equation, A1 for both values.

A second route, if you would rather not recall the derivative. Write

$$\begin{aligned} x^4+px^3-2x^2+qx-3 \\ = (x+1)^2\left(x^2+\alpha x+\beta\right) \end{aligned}$$

and equate coefficients. The constant term gives $\beta=-3$ at once, the coefficient of x^3 gives $\alpha+2=p$, and the one of x^2 gives $\beta+2\alpha+1=-2$, hence $\alpha=0$ and $p=2$. The same number of lines, and the markscheme accepts it.

About the check in the notebook. There is no reference answer here at all. Your p and q are substituted into the polynomial and `sp.div` computes the real remainder of the division by $(x+1)^2$. That is exactly the condition stated in the question, and no written-down «right answer» is needed for it.

Solution 8 — the product of the roots with a letter

For $ax^2 + bx + c = 0$ the product of the roots is $\frac{c}{a}$. Here $a = k$ and $c = 2k + 9$, so

$$\alpha\beta = \frac{2k + 9}{k}.$$

```
my8 = (2*k + 9)/k
```

The marking. A1. One mark, and the question is the first part of a long task — a gift that must not be left on the table.

What comes next. Such parts almost always lead to a condition like «the product of the roots is 4» or «the roots are reciprocal». Then $\frac{2k+9}{k}$ is equated to something and k is found — with not a single root.

A small thing. The answer can also be written as $2 + \frac{9}{k}$; the check compares values and will take either form. So will an examiner.

Solution 9 — an identity and a new equation

(a) Expand the cube:

$$(p+q)^3 = p^3 + 3p^2q + 3pq^2 + q^3.$$

Expand the second term:

$$3pq(p+q) = 3p^2q + 3pq^2.$$

Subtract — the middle terms cancel:

$$(p+q)^3 - 3pq(p+q) = p^3 + q^3. \qquad \blacksquare$$

(b) By Vieta for $2x^2 - 5x + 1 = 0$:

$$\alpha + \beta = \frac{5}{2}, \qquad \alpha\beta = \frac{1}{2}.$$

The new roots are $\frac{1}{\alpha^3}$ and $\frac{1}{\beta^3}$. Their product:

$$n = \frac{1}{(\alpha\beta)^3} = \frac{1}{(1/2)^3} = 8.$$

Their sum:

$$\frac{1}{\alpha^3} + \frac{1}{\beta^3} = \frac{\alpha^3 + \beta^3}{(\alpha\beta)^3}.$$

The numerator comes from the identity in part (a):

$$\begin{aligned}\alpha^3 &\\ &+\beta^3 &&= \left(\frac{5}{2}\right)^3\\ &-3\cdot\frac{1}{2}\cdot\frac{5}{2}\\ &= \frac{125}{8} - \frac{15}{4}\\ &= \frac{125-30}{8}\\ &= \frac{95}{8}.\end{aligned}$$

So the sum of the new roots is $\frac{95/8}{1/8} = 95$. In the equation $x^2 + mx + n = 0$ the sum of the roots is $-m$, so

$$m = -95, \qquad n = 8. \qquad \blacksquare$$

```
my9_a = p**3 + 3*p**2*q + 3*p*q**2 + q**3 -
(3*p**2*q + 3*p*q**2)
my9_m = -95
my9_n = 8
```

The marking. (a): M1 for expanding the cube, A1 for the result. (b): M1 for Vieta on the original equation, A1 for $n = 8$, M1 for expressing the sum through $\alpha^3 + \beta^3$, M1 for applying part (a), A1 for $\frac{95}{8}$, A1 for $m = -95$.

Why part (a) comes first. It is the tool for (b). Almost every two-part question in this topic is built that way: the first part prepares an identity, the second applies it. On seeing «hence» or «use the result from part (a)», do not start again — substitute.

The sign of m. The most frequent lost mark: finding the sum 95 and writing $m = 95$. In $x^2 + mx + n$ the sum of the roots is $-m$, not m. You can check on any familiar equation: $x^2 - 3x + 2$ has roots 1 and 2, the sum is 3, and the coefficient is -3.

About the check. `verify_root_transform` stores neither m nor n. It takes the roots of the original equation numerically, cubes them, inverts them and compares with the roots of yours. So the equation $8x^2 - 760x + 64 = 0$ — the same one multiplied by 8 — passes too: its roots are the same.

Solution 10 — a relation between the roots is given

The roots: α, β, $\alpha+\beta$. Write Vieta for $x^3 - kx^2 + \frac{k^2}{2}x - 3k = 0$.

The sum of the roots is k:

$$\begin{aligned} &\alpha \\ &\quad + \beta \\ &\quad + (\alpha+\beta) \quad = 2(\alpha+\beta) \\ &= k \\ &\implies \alpha+\beta \\ &= \frac{k}{2}. \end{aligned}$$

The sum of the pairwise products is $\frac{k^2}{2}$:

$$\begin{aligned} &\alpha\beta \\ &\quad + \alpha(\alpha+\beta) \\ &\quad + \beta(\alpha+\beta) \quad = \alpha\beta \\ &\quad + (\alpha+\beta)^2 \\ &= \frac{k^2}{2}. \end{aligned}$$

Substitute $(\alpha+\beta)^2 = \frac{k^2}{4}$:

$$\alpha\beta = \frac{k^2}{2} - \frac{k^2}{4} = \frac{k^2}{4}.$$

The product of the roots is $3k$:

$$\alpha\beta(\alpha+\beta) = \frac{k^2}{4}\cdot\frac{k}{2} = \frac{k^3}{8} = 3k.$$

With $k > 0$ divide by k:

$$\frac{k^2}{8} = 3 \implies k^2 = 24 \implies k = 2\sqrt{6}. \qquad \blacksquare$$

```
my10_ab = k**2/4
my10_k = 2*sqrt(6)
```

The marking. M1 for the sum of the roots, A1 for $\alpha+\beta=\frac{k}{2}$, M1 for the product of the roots, M1 for the substitution, A1 for $k=2\sqrt{6}$.

About the question. In the archive's tags the equation is written as $x^3 - kx^2 + 3k = 0$, and with that equation the answer $2\sqrt{6}$ does not exist: from $\frac{k^3}{8} = -3k$ one gets $k^2 = -24$. The constant term has to give the product of the roots as $+3k$, and the middle coefficient has to agree with $\alpha\beta = \frac{k^2}{4}$; the equation above does both and leads exactly to the answer recorded in the archive. The details are in the technique card, section `corpus_issues`.

What the task teaches. The condition «the third root equals the sum of the other two» gives nothing by itself — it starts working only when it is substituted into all three of Vieta's formulas at once. One formula is never enough: there are three unknowns here (α, β, k).

Solution 11 — find the parameter and all the roots

We are given that $z - 3i$ is a factor, so $3i$ is a root. The coefficients of P are real (for real m), so $-3i$ is a root too. Let the third root be ρ.

The product of the roots for $z^3 - \frac{36}{m}z^2 + mz - 4m$ is $4m$:

$$(3i)(-3i)\rho = 9\rho = 4m.$$

The sum of the roots is $\frac{36}{m}$:

$$3i + (-3i) + \rho = \rho = \frac{36}{m}.$$

Substitute the second into the first:

$$\begin{aligned} 9 \cdot \frac{36}{m} &= 4m \\ &\Longrightarrow\ 324 \\ &= 4m^2 \\ &\Longrightarrow\ m^2 \\ &= 81 \\ &\Longrightarrow\ m \\ &= 9 \end{aligned}$$

(we take the positive root because $m > 0$). Then $\rho = \frac{36}{9} = 4$, and the roots of P are $3i$, $-3i$, 4. ■

```
my11_m = 9
my11_roots = [3*I, -3*I, 4]
```

The marking. A1 for «$3i$ is a root», A1 for «$-3i$ is a root too», M1 for attempting to equate the sum or the product, (A1) for $m = 9$, A1 for the third root 4. A separate note in the markscheme: with no explicit mention of the roots the maximum is A0A0(M1)(M1)(A1)A1 — that is, two marks are lost simply because it was not said which numbers are taken to be roots.

A check with the middle coefficient. The sum of the pairwise products is m:

$$(3i)(-3i) + (3i + (-3i))\,\rho = 9 + 0 = 9.$$

And indeed $m = 9$. That is a spare equation, of which one would have been enough — a useful free check.

The second route. Write $P(z) = (z - 3i)(z + 3i)(z - \rho) = (z^2 + 9)(z - \rho)$, expand to $z^3 - \rho z^2 + 9z - 9\rho$ and compare the coefficients with the original. The same thing comes out; the markscheme accepts it too. Note that the pair of conjugate roots gave a real factor $z^2 + 9$ straight away — no complex numbers appear after that.

Solution 12 — an equation for the reciprocal roots

Let $w = \frac{1}{z}$, that is, $z = \frac{1}{w}$. Substitute into the original equation:

$$\frac{1}{w^4} + \frac{2}{w^2} + 4 = 0.$$

Multiply by w^4 (legitimate: $w = 0$ cannot be a root, because z is finite):

$$1 + 2w^2 + 4w^4 = 0 \implies 4w^4 + 2w^2 + 1 = 0.$$

So $p = 4$, $q = 2$, $r = 1$. ■

```
my12_p, my12_q, my12_r = 4, 2, 1
```

The marking. M1 for the substitution $w = \frac{1}{z}$, M1 for multiplying by w^4, A1 for the triple of coefficients.

Why substitution and not Vieta. There are four roots here, and computing $\sum \frac{1}{z_i}$, $\sum_{i<j} \frac{1}{z_i z_j}$ and so on means four symmetric sums instead of two lines. The rule is simple: if the new root is expressed through the old one **explicitly and invertibly**, go by substitution. Vieta is needed when the inversion does not work out or the transformation is awkwardly non-linear — like $\frac{1}{\alpha^3}$ in task 9.

A pretty consequence. The coefficients of the new equation are the coefficients of the old one the other way round: $1, 2, 4 \to 4, 2, 1$. That always happens under

$z \to \frac{1}{z}$, and it is a handy check on the answer.

Solution 13 — the Paper 3 investigation, the cubic case

(b)(i) Square the sum of the roots:

$$\begin{aligned}&(\alpha+\beta+\gamma)^2\\&=\alpha^2\\&\quad+\beta^2\\&\quad+\gamma^2\\&\quad+2(\alpha\beta+\beta\gamma+\gamma\alpha).\end{aligned}$$

By Vieta $\alpha+\beta+\gamma=-p$ and $\alpha\beta+\beta\gamma+\gamma\alpha=q$, so $p^2=\left(\alpha^2+\beta^2+\gamma^2\right)+2q$ and

$$\alpha^2+\beta^2+\gamma^2=p^2-2q. \qquad \blacksquare$$

(b)(ii) Expand the sum of the squares of the differences:

$$\begin{aligned}&(\alpha-\beta)^2\\&\quad+(\beta-\gamma)^2\\&\quad+(\gamma-\alpha)^2\\&\quad=2\left(\alpha^2+\beta^2+\gamma^2\right)-2(\alpha\beta+\beta\gamma+\gamma\alpha).\end{aligned}$$

Substitute the result of (b)(i) and q:

$$=2\left(p^2-2q\right)-2q=2p^2-6q. \qquad \blacksquare$$

(c) Suppose $p^2 < 3q$. Then $2p^2 - 6q < 0$, that is, the sum of the squares of the differences is negative. If all three roots were real, every square would be non-negative and so would the sum. A contradiction: the roots are not all real. ■

(d) For $x^3 - 7x^2 + 17x + 1 = 0$ we have $p = -7$, $q = 17$:

$$p^2 = 49, \qquad 3q = 51, \qquad 49 < 51.$$

By part (c) at least one root is not real. ■

(e)(i) The condition $p^2 \geq 3q$ at $p = -7$ gives $q \leq 16$. Run through the integers q from 1 to 16 on the calculator: up to $q = 11$ all three roots are real, and at $q = 12$ only one is. Answer: $q = 12$.

(e)(ii) For a cubic with real coefficients the non-real roots come in conjugate pairs, so there are 0 or 2 of them, and at least one root is always real.

```
my13_sq = p**2 - 2*q
my13_dif = 2*p**2 - 6*q
my13_p2, my13_3q = 49, 51
my13_q = 12
```

The marking. (b)(i): M1 for squaring the sum, A1 for the expansion, A1 for the result. (b)(ii): M1M1A1 likewise. (c): M1 for $2p^2 - 6q < 0$, R1 for the conclusion. (d): M1 for computing both numbers, A1 for the comparison and the conclusion. (e)(i): M1 for the search, A1 for $q = 12$. (e)(ii): R1.

The main idea of the question. Statement (c) is an implication one way. In part (e) Noah believes the converse holds too: «if $p^2 \geq 3q$ then all the roots are real». That is false, and $q = 12$ is a counterexample: $49 \geq 36$ and the roots are not all real. Realising that a counterexample is what is wanted is half the solution; after that the calculator does the work.

Why the converse could not have been true. The condition $2p^2 - 6q \geq 0$ says only that the *sum* of the squares of the differences is non-negative. For complex numbers a square can be negative, and the terms are able to

compensate for each other. A one-sided criterion is an ordinary thing: it catches only the obvious cases.

Where the calculator is here. Part (e) (i) is the only place in the topic where a GDC does the work rather than checks it. Running through sixteen values by hand would have taken all the time left.

Solution 14 — the Paper 3 investigation, the fourth degree

(f)(i) The same identity, only with four terms:

$$(\alpha+\beta+\gamma+\delta)^2 = \sum \alpha_i^2 + 2\sum_{i<j}\alpha_i\alpha_j.$$

By Vieta the left-hand side is p^2 and the second sum is q, so

$$\alpha^2+\beta^2+\gamma^2+\delta^2 = p^2 - 2q. \qquad \blacksquare$$

(f)(ii) If $p^2 < 2q$, the sum of the squares is negative, which is impossible for real numbers. The condition: $p^2 < 2q$.

(g) For $x^4 - 2x^3 + 3x^2 - 4x + 5$: $p = -2$, $q = 3$, so $p^2 = 4$ and $2q = 6$. Since $4 < 6$, at least one root is not real. ■

(h)(i) For $x^4 - 9x^3 + 24x^2 + 22x - 12$: $p = -9$, $q = 24$, so $p^2 = 81$ and $2q = 48$. Here $81 > 48$ and the criterion is silent — nothing can be concluded from it.

(h)(ii) The candidates for integer roots are the divisors of the constant term: $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$. Substitute $x = -1$:

$$1 + 9 + 24 - 22 - 12 = 0.$$

The integer root is -1.

(h)(iii) Divide by $(x + 1)$; the remainder is zero and the quotient is

$$x^3 - 10x^2 + 34x - 12.$$

For that cubic $p = -10$, $q = 34$, so $p^2 = 100$, $3q = 102$. Since $100 < 102$, by part (c) the cubic factor has a non-real root — and therefore so does the quartic. ■

```
my14_sq = p**2 - 2*q
my14_p2, my14_2q = 4, 6
my14_root = -1
my14_quo, my14_rem = x**3 - 10*x**2 + 34*x -
12, 0
my14_pc2, my14_3qc = 100, 102
```

The marking. (f)(i): M1A1A1. (f)(ii): A1. (g): A1. (h)(i): A1 for the conclusion «nothing can be said». (h)(ii): A1. (h)(iii): M1 for the factorisation, A1 for the cubic factor, M1 for applying the criterion, A1 for the conclusion.

A two instead of a three. For a cubic the criterion is $p^2 < 3q$, for a quartic $p^2 < 2q$. The number does not come out of nowhere: in the cubic case the sum of the squares of the **differences** was compared (which brought the factor 2 and the coefficient 6), while in the fourth degree it is the sum of the squares of the roots directly, $p^2 - 2q$. Different identities, different constants. Substituting the wrong number is the most frequent mistake in this question.

Part (h)(i) is worth a mark for the word «nothing». The answer «all the

roots are real» scores zero: the criterion is one-sided, and at $p^2 > 2q$ it asserts nothing. Being able to say «that does not follow» is an exam skill of its own.

Technique 1 inside Paper 3. Part (h) is the ordinary factor theorem and division inserted in the middle of an investigation. The whole of Paper 3 is built that way: a long storyline made of short standard steps.

Solution 15 — two decompositions in one line

(a) Write the identity and multiply by $x(k-x)$:

$$\begin{aligned} \frac{1}{x(k-x)} &= \frac{a}{x} + \frac{b}{k-x} \\ &\Longrightarrow\ 1 \\ &= a(k-x) + bx. \end{aligned}$$

Put $x = 0$: $1 = ak$, hence $a = \frac{1}{k}$. Put $x = k$: $1 = bk$, hence $b = \frac{1}{k}$.

$$\frac{1}{x(k-x)} = \frac{1}{kx} + \frac{1}{k(k-x)}.$$

(b) The denominator factorises: $1 - 4v^2 = (1 - 2v)(1 + 2v)$. Multiply:

$$1 = A(1 + 2v) + B(1 - 2v).$$

Put $v = \frac{1}{2}$: $1 = 2A$, hence $A = \frac{1}{2}$. Put $v = -\frac{1}{2}$: $1 = 2B$, hence $B = \frac{1}{2}$.

$$\frac{1}{1 - 4v^2} = \frac{1}{2(1 - 2v)} + \frac{1}{2(1 + 2v)}.$$

```
my15_a = 1/(k*x) + 1/(k*(k - x))
my15_b = Rational(1, 2)/(1 - 2*v) +
Rational(1, 2)/(1 + 2*v)
```

The marking. (a): M1 for the identity after multiplying, M1 for the substitution or for equating coefficients, A1 for both values. (b): M1A1.

Why substitution is faster. Equating coefficients in (a) gives the system $-a + b = 0$ and $ak = 1$ — not hard either, but two steps instead of one. Substituting $x = 0$ and $x = k$ kills one term at a time. The rule: take the values that make the denominators vanish.

Where this leads. Both blocks are the first parts of integration questions. In the

May 2021 block $\frac{1}{x(k-x)}$ is integrated next — the logistic equation from E7; in the November 2025 block $\frac{1}{1-4v^2}$ in a problem about motion with resistance. Partial fractions are almost never the goal; they are the first step.

Solution 16 — a repeated factor in the denominator

(c) Write the identity and multiply by $Q(x) = (x+1)(2x+1)$:

$$1 = A(2x+1) + B(x+1).$$

Put $x = -1$: $1 = -A$, hence $A = -1$. Put $x = -\frac{1}{2}$: $1 = \frac{B}{2}$, hence $B = 2$.

$$\frac{1}{Q(x)} = -\frac{1}{x+1} + \frac{2}{2x+1}.$$

(d) Notice that $(1+x)(2x+1)$ is the same $Q(x)$. So the expression equals

$$\frac{1 - \frac{x}{Q}}{Q} = \frac{Q - x}{Q^2}.$$

Compute the numerator: $Q - x = 2x^2 + 3x + 1 - x = 2x^2 + 2x + 1$. The denominator is $Q^2 = (x+1)^2(2x+1)^2$. Decompose into

partial fractions; each repeated factor gives two terms:

$$\begin{aligned}\frac{2x^2+2x+1}{(x+1)^2(2x+1)^2} &= \frac{2}{x+1} \\ &\quad + \frac{1}{(x+1)^2} \\ &\quad - \frac{4}{2x+1} \\ &\quad + \frac{2}{(2x+1)^2}.\end{aligned}$$

```
my16_c = -1/(x + 1) + 2/(2*x + 1)
my16_d = 2/(x + 1) + 1/(x + 1)**2 - 4/(2*x +
1) + 2/(2*x + 1)**2
```

The marking. (c): M1 for the identity, M1 for the substitution or the system, A1 for A and B. (d): M1 for using the result of (c), A1 for the outcome.

A fast route to (d). From (c) we know $\frac{1}{Q} = -\frac{1}{x+1} + \frac{2}{2x+1}$. Then $\frac{Q-x}{Q^2} = \frac{1}{Q} - \frac{x}{Q^2}$, and all that is left is squaring the decomposition already known. That is how part (d) really is solved «hence», in

two lines — which is what (c) stands before it for.

About repeated factors. For $(x+1)^2$ in the denominator **both** terms are needed, $\frac{A}{x+1}$ and $\frac{B}{(x+1)^2}$. If only the second is written, the system turns out inconsistent — and that is a reliable sign of an omission, not a reason to hunt for an arithmetic slip.

About the archive's tags. The right-hand side of this part is recorded there as $1-\frac{4}{2x+1}-\frac{2}{x+1}+\frac{2}{(2x+1)^2}$. Checking in sympy shows that this is not equal to the left-hand side: $\frac{2}{x+1}$ should carry a plus, and instead of the free one there should be $\frac{1}{(x+1)^2}$. The working is in the technique card, section `corpus_issues`.

Solution 17 — the task on the timer

(a) For $g(x)=2x^3-7x^2+dx-e$ the sum of the roots is $-\frac{-7}{2}$:

$$\alpha+\beta+\gamma=\frac{7}{2}.$$

(b) The coefficients of h are real, so together with $\rho+3i$ the number $\rho-3i$ is a root as well. In all h has five roots: $\alpha,\beta,\gamma,\rho+3i,\rho-3i$. Their sum is $-\frac{-11}{2}=\frac{11}{2}$:

$$\begin{aligned}
&\frac{7}{2}\\
&\qquad+(\rho+3i)\\
&\qquad+(\rho-3i) \quad=\frac{11}{2}\\
\Longrightarrow\ &\frac{7}{2}+2\rho\\
=\ &\frac{11}{2}\\
\Longrightarrow\ &\rho\\
=\ &1. \qquad \blacksquare
\end{aligned}$$

(c) The product of the five roots is $-\frac{-20}{2}=10$. From $h\left(\frac12\right)=0$ and the condition «γ is rational» it follows that $\gamma=\frac12$. The conjugate pair gives

$$(1+3i)(1-3i)=1+9=10.$$

So

$$10\cdot\alpha\beta\cdot\frac12=10\ \Longrightarrow\ \alpha\beta=2.$$

From (a) $\alpha+\beta=\frac72-\frac12=3$. Numbers with sum 3 and product 2 are the roots of

$u^2 - 3u + 2 = 0$, that is, 1 and 2. Since $\alpha < \beta$, we get $\alpha = 1$, $\beta = 2$. ■

```
my17_sum = Rational(7, 2)
my17_rho = 1
my17_ab = 2
my17_alpha, my17_beta = 1, 2
```

The marking. (a): A1. (b): A1 for the conjugate root, M1 for the sum of the five roots, A1 for $\rho = 1$. (c): M1 for the product of the five roots, A1 for $\alpha\beta = 2$, A1 for the pair of values.

How it looks in time. Seven marks in twelve minutes, and almost all the time goes on two careful applications of Vieta to a fifth degree. Not one root is found by honestly solving an equation — nor could it be: h is of the fifth degree with three letters.

The key move in (c). Noticing that $\frac{1}{2}$ is γ. The question says «γ is rational», while α, β are positive and real; the root $\frac{1}{2}$ from $h\left(\frac{1}{2}\right) = 0$ has to be one of the three, and the rationality points at γ. Without that step the product cannot be split up.

A check on the answer. $\alpha+\beta+\gamma = 1+2+\frac{1}{2} = \frac{7}{2}$ — it agrees with part (a). Such a check takes five seconds and catches almost any arithmetic slip in (c).